

Brayden Letwin

Mathematics • Theoretical Computer Science

BSc Mathematics, University of Alberta • PhD Mathematics, University of Washington (from 2026)

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ABOUT

I am completing a BSc in Mathematics at the University of Alberta and will enter the PhD program in Mathematics at the University of Washington in September 2026 under Dan Mikulincer. My interests include convex geometry, random matrices, probability, theoretical computer science, and related high-dimensional phenomena. I am currently working under the supervision of Alexander Litvak and Beatrice-Helen Vritsiou.

EDUCATION

2026–present	University of Washington PhD in Mathematics Incoming PhD student.
2022–2026	University of Alberta Bachelor of Science (Honours), Mathematics; GPA 3.7/4.0

SELECTED PUBLICATIONS AND PREPRINTS

- **B. Letwin and C. Tang.** *How anisotropic can the sections of an isotropic convex body be?* In preparation.
- **A. R. Polavarapu and B. Letwin.** *A lower bound for the smallest singular value of a weighted random matrix.* In preparation.
- **B. Letwin and D. Mikulincer** *Dimension free Gaussian tail estimates for linear functionals on convex bodies* In preparation.
- **B. Letwin and M. Sawhney.** *On the maxima of Littlewood polynomials on $[-1, 1]$.* [arXiv:2604.19294](https://arxiv.org/abs/2604.19294).
- **B. Letwin and V. Yaskin.** *A generalization of Grünbaum's inequality.* *Israel Journal of Mathematics.* [arXiv:2410.04741](https://arxiv.org/abs/2410.04741).

RESEARCH EXPERIENCE

Summer 2026	NSERC USRA Supervisor: Beatrice-Helen Vritsiou Studying the mean-width of an isotropic convex body.
Summer 2025	MSS USRA Supervisors: Alexander Litvak and Beatrice-Helen Vritsiou Studied covering numbers of convex bodies and related quantities.
Summer 2023; Summer 2024	NSERC USRA Supervisor: Vladyslav Yaskin Studied variations of Grünbaum-type inequalities and their applications.

TEACHING EXPERIENCE

Over the past two years I have held fifteen teaching-assistant appointments, typically assisting three classes per semester. Duties have included seminars, grading, office hours, and proctoring.

- **University of Alberta** - MATH 216 (Introduction to Analysis), 2024–2025.
- **University of Alberta** - CMPUT 304 (Algorithms II), 2024–2025.
- **University of Alberta** - MATH 146 (Calculus for the Mathematical and Physical Sciences II), 2023–2024.
- **University of Alberta** - MATH 144 (Calculus for the Mathematical and Physical Sciences I), Winter 2024.
- **University of Alberta** - CMPUT 175 (Introduction to the Foundations of Computation II), 2023–2025.

TALKS AND LECTURES

October 2025	High Dimensional Probability Seminar, Texas A&M University College Station, TX, USA Talk: <i>On a generalization of Grünbaum's inequality.</i>
September– December 2025	CMPUT 304 Seminar Lectures, University of Alberta Edmonton, AB, Canada Talk: <i>On linear programs and their applications.</i>
May 2024	Approximation Algorithms Seminar, University of Alberta Edmonton, AB, Canada Talk: <i>On a 4-approximation algorithm for the weak chromatic rectilinear art-gallery problem.</i>
November 2023	CMS Winter Meeting: Geometric Functional Analysis Montreal, QC, Canada Talk: <i>On a generalization of Grünbaum's inequality.</i>

AWARDS AND DISTINCTIONS

NebuLogic Technologies Award, University of Alberta	2025
MSS Undergraduate Student Research Award, University of Alberta	2025
Cyril G Wates Memorial Scholarship, University of Alberta	2024
NSERC Undergraduate Student Research Award, University of Alberta	2024
Jason Lang Scholarship, University of Alberta	2023
Murray Thomas Gibson Memorial Scholarship in Mathematics, University of Alberta	2023
Dean's Honor Roll, University of Alberta	2023
NSERC Undergraduate Student Research Award, University of Alberta	2023
Max Wyman Memorial Entrance Leadership Scholarship, University of Alberta	2021
Alexander Rutherford Scholarship, University of Alberta	2021

PROJECTS AND COMPETITIONS

IMC Prosperity 3 Trading Challenge - finished top 1%	<i>April 2025</i>
Rotman International Trading Competition - finished top 30%	<i>February 2024</i>

WORKSHOPS/CONFERENCES ATTENDED

High-dimensional Probability and its Applications, Evanston, IL, USA	<i>August 2026</i>
Synergies between Geometry, Probability, and Computation in High Dimensions, Providence, RI, USA	<i>July 2026</i>
PAWS: Symmetries of Root Systems, Online	<i>October 2024</i>
CMS Winter Meeting: Geometric Functional Analysis, Montreal, QC, Canada	<i>November 2023</i>
Harmonic Analysis and Convexity, Banff International Research Station, Banff, AB, Canada	<i>November 2023</i>

ACADEMIC SERVICE AND VOLUNTEER WORK

2024–2025	Undergraduate Mathematical Sciences Society, University of Alberta Edmonton, AB, Canada Society executive.
July 2024	CMS Alberta Summer Mathematics Camp, University of Alberta Edmonton, AB, Canada Lead coordinator.
2022–2023	Undergraduate Mathematical Sciences Society, University of Alberta Edmonton, AB, Canada Society executive.